Gas laws

KINETIC MOLECULAR THEORY OF GASES

Maxwell and Boltzmann (1859) developed a mathematical theory to explain the behaviour of gases and the gas laws. It is based on the fundamental concept that **a gas is made of a large number of molecules in perpetual motion.** Hence the theory is called the **kinetic molecular theory** or simply the **kinetic theory of gases** (The word kinetic implies motion). The kinetic theory makes the following assumptions.

Assumptions of the Kinetic Molecular Theory

(1) **A gas consists of extremely small discrete particles called molecules dispersed throughout the container.** The actual volume of the molecules is negligible compared to the total volume of the gas. The molecules of a given gas are identical and have the same mass (m).

(2) **Gas molecules are in constant random motion with high velocities.** They move in straight lines with uniform velocity and change direction on collision with other molecules or the walls of the container.

(3) The distance between the molecules are very large and it is assumed that van der Waals attractive forces between them do not exist. Thus **the gas molecules can move freely, independent of each other.**

(4) All collisions are perfectly elastic. Hence, **there is no loss of the kinetic energy of a molecule during a collision.**

(5) **The pressure of a gas is caused by the hits recorded by molecules on the walls of the container.**

(6) The average kinetic energy ($1/2\ mv^2$) of the gas molecules is directly proportional to absolute temperature (Kelvin temperature). This implies that **the average kinetic energy of molecules is the same at a given temperature.**

1.14 Ideal and Real gases

By the application of gas laws (molar gas equation) gases can be classified into two classes :

(i) ideal gas and (ii) real gas

Ideal gas : The gas which obeys the molar gas equation, $PV = nRT$ (i.e., Boyle's law, Charles' law, Avogadro's law) at all temperatures and pressures is called **ideal gas**. It should also follow the equation, $\left(\frac{\delta E}{\delta V}\right)_T = 0$, where δE = change of internal energy, δV = change in volume). It means at constant temperature, the change of volume of a gas does not produce any change of internal energy of the gas. It is an imaginary concept in reality there is no example of ideal gas. A plot of PV vs. p should give straight horizontal line, but in reality it is not found at all.

Real gase : The gas which does not follow gas laws at any temperature and pressure is known as **real gas.** Hydrogen, oxygen and nitrogen etc. do not follow Boyle's and Charles' law at ordinary temp. and presence. So, these gases are real gases. At extremely low pressure and high temperature some gases particularly the noble gases behave like an ideal gas.

The differences of ideal and real gas are given in Table 1.4:

Table 1.4 : Difference of ideal and real gas

	Ideal Gas		Real Gase
1	The gas which follows "gas laws" such as Boyle's, Charles' and Avogadro's law is known as "ideal gas".	1	The gas, which does not follow gas laws at any temperature and pressure, is known as "real gas".
2	Intermolecular attraction force of ideal gas is negligible.	2	Intermolecular force of real gas is considerable but not negligible.
3	The volume of the molecules is negligible compared to the volume of the container.	3	Volume of the gas molecule is considerable but not negligible.
4	Ideal gas follows gas laws at any temperature and pressure.	4	Real gas follows gas laws at high temperature and low pressure.
5	Ideal gas is an imaginary concept. There is no example of ideal gas.	5	There are many examples of real gas such as H_2, N_2, CO_2 etc.

DEDUCTION OF GAS LAWS FROM THE KINETIC GAS EQUATION

(*a*) Boyle's Law

According to the Kinetic Theory, there is a direct proportionality between absolute temperature and average kinetic energy of the molecules *i.e.*,

$$\frac{1}{2}mNu^2 \propto T$$

or $$\frac{1}{2}mNu^2 = kT$$

or $$\frac{3}{2} \times \frac{1}{3}mNu^2 = kT$$

or $$\frac{1}{3}mNu^2 = \frac{2}{3}kT$$

Substituting the above value in the kinetic gas equation $PV = \frac{1}{3}mNu^2$, we have

$$PV = \frac{2}{3}kT$$

The product PV, therefore, will have a constant value at a constant temperature. This is Boyle's Law.

(*b*) Charles' Law

As derived above,

$$PV = \frac{2}{3}kT$$

or $$V = \frac{2}{3} \times \frac{k}{P}T$$

At constant pressure,

$$V = k'\,T \qquad \text{where } \left(k' = \frac{2}{3} \times \frac{k}{P}\right)$$

or $$V \propto T$$

That is, at constant pressure, volume of a gas is proportional to Kelvin temperature and this is Charles' Law.

(*c*) Avogadro's Law

If equal volume of two gases be considered at the same pressure,

$$PV = \frac{1}{3}m_1N_1u_1^2 \qquad \text{...Kinetic equation as applied to one gas}$$

$$PV = \frac{1}{3}m_2N_2u_2^2 \qquad \text{...Kinetic equation as applied to 2nd gas}$$

$\therefore$ $$\frac{1}{3}m_1N_1u_1^2 = \frac{1}{3}m_2N_2u_2^2 \qquad ...(1)$$

When the temperature (T) of both the gases is the same, their mean kinetic energy per molecule will also be the same.

i.e., $$\frac{1}{3}m_1u_1^2 = \frac{1}{3}m_2u_2^2 \qquad ...(2)$$

Dividing (1) by (2), we have

$$N_1 = N_2$$

Or, under the same conditions of temperature and pressure, equal volumes of the two gases contain the same number of molecules. This is Avogadro's Law.

(*d*) Graham's Law of Diffusion

If m_1 and m_2 are the masses and u_1 and u_2 the velocities of the molecules of gases 1 and 2, then at the same pressure and volume

$$\frac{1}{3} m_1 N_1 u_1^2 = \frac{1}{3} m_2 N_2 u_2^2$$

By Avogadro's Law $\quad N_1 = N_2$

$$\therefore \quad m_1 u_1^2 = m_2 u_2^2$$

or
$$\left(\frac{u_1}{u_2}\right)^2 = \frac{m_2}{m_1}$$

If M_1 and M_2 represent the molecular masses of gases 1 and 2,

$$\left(\frac{u_1}{u_2}\right)^2 = \frac{M_2}{M_1}$$

$$\frac{u_1}{u_2} = \sqrt{\frac{M_2}{M_1}}$$

The rate of diffusion (r) is proportional to the velocity of molecules (u), Therefore,

$$\frac{\text{Rate of diffusion of gas 1}}{\text{Rate of diffusion of gas 2}} = \frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

This is Graham's Law of Diffusion.

Problem: Calculate the root mean square velocity of an oxygen molecule at 27° C.

We know that, $C = \sqrt{\frac{3RT}{M}}$

$= \sqrt{\frac{3 \times 8.316 \times 10^7 \times 300}{32}}$

$= 4.836 \times 10^4 \text{ cms}^{-1}$

$= 483 \text{ ms}^{-1}$

Here,

Temperature, T = 27+273=300 K

Molar mass of O_2, M=32

Gas constant value = $8.316 \times 10^7 \text{ ergK}^{-1}\text{mol}^{-1}$

Root mean square velocity, C = ?

Establish the ideal gas equation $PV = nRT$

Boyle's law states that the pressure of a given mass of an ideal gas is inversely proportional to its volume at a constant temperature.
We can write $\alpha \frac{1}{P}$, Where V for volume and P for pressure

Charle's law states that the volume of an ideal gas at constant pressure is directly proportional to the absolute temperature.
We can write $V \alpha T$, where T for temperature and

The Avogadro's law stating that equal volumes of gases at the same temperature and pressure contain equal numbers of molecules.
We can write $V \alpha\, n$, where n for mole number

When we combine the above three laws $V \alpha \frac{n}{P} \times T$

$PV \alpha RT$ or $PV = nRT$, where R is gas constant.

This equation is called ideal gas equation.

Find the value of *R* where *R* is a gas constant

We know the ideal gas equation is $PV = nRT$

Where P is pressure, V is volume, n is number of moles, R is Universal Gas Constant, T is temperature. At standard temperature and pressure (273.15 K and 1 bar), 1 mol of gas occupies 22.711 L.

In SI units, then pressure is measured in pascals and volume is measured in cubic metres.

$$R = \frac{PV}{nT} = \frac{1.013325 \times 10^5 \text{ Pa} \times 22.414 \times 10^{-3}\, m^3}{1\, mol \times 273.16\, K}$$

$= 8.3145$ Pa·m^{3}K^{-1}mol^{-1}

Always make sure that you use the value of R corresponding to the units that you are using for P and V.